



Requirements Traceability Matrix (RTM)

TASCO Insurance × VETC Growth Platform — Business & Product

iorta TechNXT

Version 1.0 · 07 October 2026

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Document	Version	Date	Owner	Status
Requirements Traceability Matrix	1.0	07 October 2026	iorta TechNXT — Business Analysis and QA	Issued for submission

TASCO Insurance × VETC Motor Insurance Growth Platform · prepared by iorta TechNXT

0. How to use this matrix

- Each row traces one functional requirement (doc 02) to its user stories (doc 04), code modules, HTTP endpoints, rule kinds and tests. References such as "doc 02" point to the companion documents listed in the Business and Product Documentation index (00-index.pdf).
- Code paths** are relative to the repository root. `domain/` = `src/domain/`, `app/` = `src/application/`, `http/` = `src/adapters/http/`, `persist/` = `src/adapters/persistence/`, `integ/` = `src/adapters/integrations/`, `rules/` = `src/rules/`, `shared/` = `src/shared/`. All API endpoints are declared in `src/adapters/http/routes.js` (80 API routes, described in `docs/api/openapi.json`); `/health/*`, `/metrics`, `/api/openapi.json` and static pages are served by `src/adapters/http/app.js`.
- Build status** is taken from doc 02 as of 07 October 2026: B = Built, P = Partial, PI = Planned (roadmap).
- Test status.** All test files below exist and run in CI. Latest run: 238 tests, 235 pass, 0 fail, 3 to-do (manual or roadmap scenarios US-055, US-076, US-077). Coverage 99.5% lines, 91.6% branches, 97.2% functions. Detailed results are in `docs/quality/TASCO-Test-Cases-and-Results.xlsx`.

0.1 Test file legend

The "Test file(s)" column uses the short codes below. Functional tests (F-1 to F-3) contain one test per Given/When/Then scenario in doc 04, named "US-*nnn* · scenario title" (124 scenarios: 121 automated, 3 manual or roadmap).

Code	Test file	Layer and scope
U-ID	<code>test/unit/identity.test.js</code>	Unit: plate, phone and spoken-plate parsing
U-JL	<code>test/unit/jsonLogic.test.js</code>	Unit: JSON Logic, templates, decision tables
U-PF	<code>test/unit/platform.test.js</code>	Unit: crypto, JWT, TOTP, validation, resilience, config, schema drift
U-RD	<code>test/unit/rulesDomain.test.js</code>	Unit: rule validators, enrichment, leads, journeys, benefits, rating, contact policy
U-ST	<code>test/unit/stores.test.js</code>	Unit: memory store, outbox, audit hash chain
U-VB	<code>test/unit/voicebot.test.js</code>	Unit: voice bot dialogue
I-GV	<code>test/integration/governance.test.js</code>	Integration: rules, identity, access, voice, customer, ops
I-JN	<code>test/integration/journeys.test.js</code>	Integration: seed, journeys, triggers, recompute
I-SA	<code>test/integration/sales.test.js</code>	Integration: quote, purchase, compensation, partners
API	<code>test/api/api.test.js</code>	HTTP contract, authentication and authorisation
SEC	<code>test/security/security.test.js</code>	Security controls (OWASP Top 10)
PG	<code>test/pg/postgres.test.js</code>	PostgreSQL adapter, migrations, audit immutability
PERF	<code>test/perf/load.js</code>	Load smoke (p95 budget)

Code	Test file	Layer and scope
F-1	test/functional/ep01-04-data-leads-journeys.test.js	Functional: US-001 – US-022
F-2	test/functional/ep05-08-voice-sales-app-value.test.js	Functional: US-023 – US-048
F-3	test/functional/ep09-15-partners-claims-privacy-governance.test.js	Functional: US-049 – US-077

1. Matrix

A. Data and MDM

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-001	US-001	app/ingestionService.js	POST /api/data/ingest	—	F-1, API	B
FR-002	US-001	domain/identity.js#normalizePlate	POST /api/data/ingest, POST /api/partner/v1/quotes	—	U-ID, F-1	B
FR-003	US-001	domain/identity.js#normalizePhone	as above	—	U-ID, F-1	B
FR-004	US-001	domain/enrichment.js#buildProfiles, survivors	POST /api/data/ingest	enrichment	U-RD, F-1	B
FR-005	US-002	domain/enrichment.js#inferExpiry, rollForward	GET /api/customers/:id	enrichment	U-RD, F-1	B
FR-006	US-006	domain/enrichment.js#inferCategory, rules/decisionTable.js	GET /api/customers/:id	enrichment	U-RD, U-JL, F-1	B
FR-007	US-004	domain/enrichment.js, app/opsService.js#dqIssues, resolveDq, app/voiceService.js#finalize	GET /api/dq/issues, POST /api/dq/issues/:id/resolve	enrichment	U-RD, I-GV, F-1	B
FR-008	US-002	app/opsService.js#lineage	GET /api/customers/:id/lineage	—	I-GV, F-1	B
FR-009	US-003	http/routes.js (handler), domain/enrichment.js#applyDeclaredExpiry	PATCH /api/customers/:id/expiry	service_levels	API, F-1	B
FR-010	US-005, US-027	app/ingestionService.js#rebuild, app/customerService.js#declareExpiry, app/voiceService.js#finalize, app/salesService.js#purchase	POST /api/customer/expiry, POST /api/voice/sessions/:id/turns	enrichment	U-RD, I-GV, F-1	B
FR-011	US-001	app/ingestionService.js#rebuild, bootstrap/container.js#registerSubscribers	POST /api/data/ingest	—	I-JN, F-1	B

B. Lead intelligence

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-012	US-007	domain/leads.js#factsFor	—	all lead kinds	U-RD, F-1	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-013	US-017, US-018	domain/leads.js#assignJourney, rules/jsonLogic.js	GET /api/leads	journeys	U-RD, U-JL, F-1	B
FR-014	US-007	domain/leads.js#score	GET /api/leads	scoring	U-RD, F-1	B
FR-015	US-008, US-054	domain/leads.js#evaluateLead, rules/decisionTable.js	GET /api/leads	nba	U-RD, U-JL, F-1, F-3	B
FR-016	US-019	domain/leads.js#benefitsFor	GET /api/leads, GET /api/customers/:id	benefits	U-RD, F-1	B
FR-017	US-007, US-009	app/leadService.js#list, app/accessPolicy.js	GET /api/leads	abac	API, SEC, I-GV, F-1	B
FR-018	US-011	app/leadService.js#recompute, bootstrap/container.js (subscribers), jobs/cli.js (recompute)	POST /api/leads/recompute	—	I-JN, F-1	B
FR-019	US-010	app/leadService.js#recomputeOne, hasActiveTascoCover	—	—	I-SA, F-1	B

C. Journeys and messaging

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-020	US-012, US-017, US-019	domain/leads.js, rules/validators.js	—	journeys	U-RD, F-1	B
FR-021	US-012	domain/leads.js#planTouchpoints	GET /api/touchpoints	journeys	U-RD, F-1	B
FR-022	US-012, US-017	app/journeyService.js#runDue, executeTouchpoint, jobs/cli.js (journeys)	POST /api/journeys/run	journeys	I-JN, API, F-1	B
FR-023	US-013, US-015	domain/contactPolicy.js#canContact	POST /api/journeys/run, POST /api/ecosystem/events	contact_policy	U-RD, I-JN, F-1	B
FR-024	US-014, US-067	domain/contactPolicy.js#checkCopy, rules/validators.js#copyViolations	POST /api/rules, POST /api/journeys/run	copy_guard	U-RD, I-SA, F-1, F-3	B
FR-025	US-012, US-015, US-019	app/journeyService.js#sendMessage, rules/jsonLogic.js#template, bootstrap/container.js (links)	POST /api/journeys/run	content.messages	I-JN, F-1	B
FR-026	US-018, US-020, US-021, US-022	app/journeyService.js#handleEcosystemEvent	POST /api/ecosystem/events	triggers, contact_policy	I-JN, API, F-1	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-027	US-040	bootstrap/container.js (subscriber confirmation-and-cross-sell)	POST /api/customer/orders (side effect)	content.messages	I-SA, F-2	B
FR-028	US-046	bootstrap/container.js (subscriber confirmation-and-cross-sell)	—	journeys.crossSell	I-SA, F-2	B
FR-029	US-012	app/journeyService.js#schedule	GET /api/touchpoints	—	API, F-1	B
FR-030	US-010	app/journeyService.js#executeTouchpoint	POST /api/journeys/run	—	I-SA, F-1	B

D. AI voice bot

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-031	US-023	domain/voicebot.js#createDialogue.start	POST /api/voice/sessions	content.voicebot	U-VB, F-2	B
FR-032	US-024	domain/voicebot.js#turn (state verify_plate), domain/identity.js#extractPlateFromSpeech	POST /api/voice/sessions/:id/turns	content.voicebot	U-VB, U-ID, F-2	B
FR-033	US-024	domain/voicebot.js (state confirm_expiry, maskPlate)	POST /api/voice/sessions/:id/turns	content.voicebot	U-VB, F-2	B
FR-034	US-026	domain/voicebot.js#keywordClassifier	POST /api/voice/sessions/:id/turns	content.voicebot	U-VB, F-2	B
FR-035	US-025	domain/voicebot.js (intent price)	POST /api/voice/sessions/:id/turns	content.voicebot, copy_guard	U-VB, F-2	B
FR-036	US-027, US-028	app/voiceService.js#finalize	POST /api/voice/sessions/:id/turns, POST /api/voice/campaign	service_levels	U-VB, I-GV, F-2	B
FR-037	US-030	app/voiceService.js#start, turn, get	POST /api/voice/sessions, GET /api/voice/sessions/:id, POST /api/voice/sessions/:id/turns	—	I-GV, API, F-2	B
FR-038	US-029, US-054	http/routes.js (campaign handler), app/voiceService.js#canCall, autoCall, integ/simulatedCaller.js	POST /api/voice/campaign	contact_policy	I-GV, API, F-2, F-3	B
FR-039	US-031	app/insightsService.js#governance	GET /api/dashboard/governance	—	API, F-2	B

E. Telesales

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-040	US-032	domain/voicebot.js#handoffSummary	GET /api/handoffs/:id	—	U-VB, F-2	B
FR-041	US-033	app/voiceService.js#listHandoffs, app/accessPolicy.js#check	GET /api/handoffs, GET /api/handoffs/:id	abac	SEC, I-GV, F-2	B
FR-042	US-033	app/voiceService.js#updateHandoff	PATCH /api/handoffs/:id	—	API, F-2	B
FR-043	US-034	http/routes.js (handler), app/accessPolicy.js#require	PATCH /api/handoffs/:id	rbac	API, F-2	B
FR-044	US-016	app/voiceService.js#createDirectHandoff, app/journeyService.js	POST /api/journeys/run	journeys	F-1	B

F. Sales and issuance

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-045	US-035	app/salesService.js#catalogue	GET /api/products	products	I-SA, F-2	B
FR-046	US-035, US-039, US-041, US-044	app/salesService.js#quote, markSent, domain/rating.js#coverStartDate	POST /api/quotes, POST /api/quotes/:id/send, POST /api/customer/quotes, POST /api/partner/v1/quotes	products, tariff.*, rating.*, benefits, service_levels	I-SA, U-RD, F-2	B
FR-047	US-035, US-039, US-051	app/salesService.js#purchase, integ/mockGateways.js (wallet, core), shared/resilience.js	POST /api/quotes/:id/send, GET /api/customer/quotes, POST /api/customer/orders, POST /api/partner/v1/orders	—	I-SA, SEC, API, F-2, F-3	B
FR-048	US-042	app/salesService.js#purchase (compensation)	as above	—	I-SA, F-2	B
FR-049	US-039	app/salesService.js#purchase (golden-record update)	as above	—	I-SA, F-2	B
FR-050	US-040, US-077	app/salesService.js#verifyCertificate, http/app.js (/verify/* → verify.html)	GET /api/public/certificates/:certNo, /verify/:certNo	—	API, F-2, F-3 (manual QR scan)	B
FR-051	US-035	app/salesService.js#listPolicies	GET /api/policies	—	API, F-2	B
FR-052	US-074	app/opsService.js#reconcile, jobs/cli.js (reconcile)	POST /api/ops/jobs/:kind (reconciliation)	—	I-SA, F-3	B

G. Products and rating

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-053	US-006, US-039, US-041	domain/rating.js#METHODS.tariff_table	POST /api/quotes (+ customer and partner variants)	tariff.tnds_car, tariff.tnds_motorbike	U-RD, F-1, F-2	B
FR-054	US-045	domain/rating.js#METHODS.rate_on_sum_insured, app/salesService.js#recordInspection	as above; POST /api/quotes/:id/inspection	rating.motor_pd	U-RD, F-2	B
FR-055	US-044	domain/rating.js#METHODS.per_seat	as above	rating.pa_seat	U-RD, F-2	B
FR-056	US-025, US-039	domain/rating.js#rate (priceRegulated note)	as above	products	U-RD, F-2	B
FR-057	US-063	rules/validators.js (products), domain/rating.js	POST /api/rules	products	U-RD, F-3	B

H. Value and benefits

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-058	US-043	domain/leads.js#benefitsFor, app/leadService.js#benefitsForStaff	GET /api/customer/home, GET /api/customers/:id	benefits	U-RD, F-2	B
FR-059	US-043	domain/leads.js#benefitsFor	as above	benefits	U-RD, F-2	B
FR-060	US-047	— (gated config only)	—	benefits (loyalty_points), referral	F-2	PI
FR-061	US-048	—	— (roadmap)	benefits (auto_renew)	F-2 (guard only)	PI
FR-062	US-020	app/journeyService.js#handleEcosystemEvent	POST /api/ecosystem/events	triggers, benefits	I-JN, F-1	P

I. Partners and fleet

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-063	US-049, US-053	app/partnerService.js#create, setStatus, list	GET/POST /api/partners, PATCH /api/partners/:id	—	I-SA, API, F-3	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-064	US-049	app/partnerService.js#issueApiKey, revokeApiKey, authenticateKey, shared/crypto.js	POST /api/partners/:id/keys, DELETE /api/partners/keys/:keyId	—	I-SA, SEC, F-3	B
FR-065	US-050	http/routes.js (partner quote handler), app/ingestionService.js, app/salesService.js#quote	POST /api/partner/v1/quotes	products, enrichment	I-SA, API, F-3	B
FR-066	US-051	http/routes.js (ownership check), app/salesService.js#purchase	POST /api/partner/v1/orders	—	API, F-3	B
FR-067	US-052	app/salesService.js#listPolicies, app/partnerService.js#statement	GET /api/partner/v1/policies, GET /api/partner/v1/statement	—	API, F-3	B
FR-068	US-052, US-068	domain/rating.js#commissionFor, rules/validators.js (commission)	POST /api/partner/v1/orders (side effect), POST /api/rules	commission	U-RD, I-SA, F-3	B
FR-069	US-052	app/partnerService.js#statement	GET /api/partners/:id/statement	—	F-3	B
FR-070	US-054, US-055	domain/leads.js (NBA), http/routes.js (campaign excludes company)	GET /api/leads?action=route_b2b	nba, journeys, benefits	F-3 (US-055 roadmap)	P

J. Claims FNOL

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-071	US-056	app/claimsService.js#submit	POST /api/customer/claims	service_levels, content.messages	I-GV, API, F-3	B
FR-072	US-057	app/claimsService.js#transition, list (TRANSITIONS)	GET /api/claims, PATCH /api/claims/:id	—	F-3	B
FR-073	US-058	app/claimsService.js#list	GET /api/customer/claims	—	API, F-3	B

K. Customer self-service and privacy

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-074	US-036	bootstrap/container.js (links.sign/verify), app/customerService.js#issueCustomerToken	POST /api/customer/session	abac (customer_self)	API, SEC, F-2	B
FR-075	US-037	app/customerService.js#home	GET /api/customer/home	benefits	I-GV, API, F-2	B
FR-076	US-038	app/customerService.js#declareExpiry	POST /api/customer/expiry	service_levels	U-RD, I-GV, F-2	B

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-077	US-035, US-039	app/salesService.js	POST /api/customer/quotes, POST /api/customer/orders	products, tariff.*, rating.*	API, I-SA, F-2	B
FR-078	US-028, US-059	app/customerService.js#updateConsent, app/ingestionService.js#rebuild (overrides)	PUT /api/customer/consent	contact_policy	I-GV, API, F-2, F-3	B
FR-079	US-060	app/customerService.js#exportData	GET /api/customer/data-export	—	I-GV, API, F-3	B
FR-080	US-061	app/customerService.js#exportData, erase	POST /api/dsar/:id/export, POST /api/dsar/:id/erase	retention	I-GV, SEC, F-3	B
FR-081	US-062	app/accessPolicy.js#maskProfile, shared/util.js#maskPhone, maskName	GET /api/customers/:id	rbac	I-GV, SEC, F-3	B

L. Rules governance

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-082	US-063	app/rulesService.js#list, byId, get	GET /api/rules, GET /api/rules/:id	all	I-GV, F-3	B
FR-083	US-063, US-067, US-068	app/rulesService.js#createDraft, rules/validators.js, rules/jsonLogic.js#validate, rules/decisionTable.js#validateTable	POST /api/rules	all	U-RD, U-JL, I-GV, F-3	B
FR-084	US-063	app/rulesService.js#validate	POST /api/rules/validate	all	API, F-3	B
FR-085	US-064	http/routes.js (simulate handler), domain/leads.js#evaluateLead	POST /api/rules/simulate	scoring, nba, journeys, benefits	API, F-3	B
FR-086	US-065	app/rulesService.js#submit, approve, reject	POST /api/rules/:id/submit, /approve, /reject	all; rbac (restricted kinds)	I-GV, SEC, PG, F-3	B
FR-087	US-066	app/rulesService.js#rollback	POST /api/rules/:id/rollback	all	I-GV, F-3	B
FR-088	—	app/rulesService.js#loadDefaults, bootstrap/seed.js#seedRules	— (CLI npm run job rules)	all	U-RD, PG	B

M. Identity and access

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-089	US-069, US-070	app/identityService.js#login, shared/crypto.js#hashPassword, verifyPassword	POST /api/auth/login	—	I-GV, SEC, F-3	B
FR-090	US-069	app/identityService.js#verifyMfa, shared/crypto.js#totp, verifyTotp	POST /api/auth/mfa, GET /api/demo/totp/:username (demo only)	—	U-PF, I-GV, API, F-3	B
FR-091	US-069	shared/crypto.js#signJwt, verifyJwt, app/identityService.js#logout, authenticate	POST /api/auth/logout, GET /api/auth/me	—	U-PF, SEC, F-3	P
FR-092	US-071	app/accessPolicy.js#permissions, require	all staff routes	rbac	I-GV, SEC, F-3	B
FR-093	US-009, US-033	app/accessPolicy.js#check	GET /api/customers/:id, GET/PATCH /api/handoffs*	abac	I-GV, SEC, F-1, F-2	B
FR-094	US-071	app/identityService.js#createUser, update, list, reset	GET/POST /api/users, PATCH /api/users/:id, POST /api/users/:id/reset	rbac	I-GV, I-JN, API, F-3	B
FR-095	US-036, US-049	app/partnerService.js#authenticateKey, app/identityService.js#authenticate, http/app.js	partner and customer routes	—	SEC, API, F-2, F-3	B
FR-096	US-071	app/identityService.js#changePassword, shared/crypto.js#checkPasswordPolicy	POST /api/auth/password	—	U-PF, I-GV, F-3	B

N. Audit

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-097	US-003, US-062	app/auditService.js#record (called by all services), persist/memoryStore.js, persist/postgresStore.js	—	—	U-ST, I-GV, F-3	B
FR-098	US-072	app/auditService.js#list	GET /api/audit	—	API, F-3	B
FR-099	US-072	persist/auditChain.js#verifyChain, entryHash, canonical, db/migrations/001_init.sql, 002_integrity_hardening.sql (append-only triggers)	GET /api/audit/verify	—	U-ST, PG, SEC, F-3	B

O. Operations, reporting and UI

FR	US	Code module(s)	API endpoint(s)	Rule kind(s)	Test file(s)	Build
FR-100	US-073	app/insightsService.js#overview	GET /api/dashboard/overview	costs	API, F-3	B
FR-101	—	app/insightsService.js#adoption	GET /api/dashboard/adoption	—	I-GV, API	B
FR-102	US-031	app/insightsService.js#governance	GET /api/dashboard/governance	—	API, F-2	B
FR-103	US-074	http/routes.js (ops status), shared/resilience.js	GET /api/ops/status	—	API, F-3	B
FR-104	US-074	app/opsService.js#recordRun, jobRuns, adapters/messaging/outboxEventBus.js, jobs/cli.js	GET /api/ops/jobs, POST /api/ops/jobs/:kind	—	I-GV, API, F-3	B
FR-105	US-075	app/opsService.js#applyRetention	POST /api/ops/jobs/retention	retention	I-GV, F-3	P
FR-106	—	http/app.js, http/openapi.js, shared/metrics.js	/health/live, /health/ready, /metrics, /api/openapi.json, GET /api/meta	—	U-PF, API, PERF	B
FR-107	—	bootstrap/seed.js, integ/syntheticVetcSource.js, shared/config.js	GET /api/demo/totp/:username, GET /api/meta (demo customers)	—	I-JN, I-GV, SEC	B
FR-108	US-022, US-076	public/index.html, public/js/console/*	all staff routes	—	API, F-1; UAT (manual)	B
FR-109	US-077	public/app/index.html, public/js/customer/*, public/verify.html	customer and public routes	—	API, F-2; UAT (manual)	B
FR-110	US-076	public/js/shared/* (i18n, theme)	—	—	F-3 (manual); UAT	B

2. Coverage summary

2.1 Traceability coverage

Capability	FRs	With ≥ 1 user story	With API endpoint	With rule kind
A Data and MDM	11	11	11	6
B Lead intelligence	8	8	6	6
C Journeys and messaging	11	11	9	9
D Voice bot	9	9	9	7
E Telesales	5	5	5	3
F Sales and issuance	8	8	8	2
G Products and rating	5	5	5	5
H Value and benefits	5	5	3	5
I Partners and fleet	8	8	8	3
J Claims	3	3	3	1
K Customer and privacy	8	8	8	7
L Rules governance	7	6	6	7
M Identity and access	8	8	8	3
N Audit	3	3	2	0
O Ops and UI	11	8	10	2
Total	110	106	101	66

2.2 Build status

Capability	Built	Partial	Planned
A Data and MDM	11	0	0
B Lead intelligence	8	0	0
C Journeys and messaging	11	0	0
D Voice bot	9	0	0
E Telesales	5	0	0
F Sales and issuance	8	0	0
G Products and rating	5	0	0
H Value and benefits	2	1	2
I Partners and fleet	7	1	0
J Claims	3	0	0
K Customer and privacy	8	0	0
L Rules governance	7	0	0
M Identity and access	7	1	0

Capability	Built	Partial	Planned
N Audit	3	0	0
O Ops and UI	10	1	0
Total	104	4	2

Counting rule:

"Built" counts B rows, including rows whose doc 02 status names a planned production integration (for example the real VETC wallet). "Partial" counts P rows and "Planned" counts PI rows. FR-088, FR-101, FR-106, FR-107 are platform requirements without a dedicated user story.

3. Engineering notes: inconsistencies found during documentation

These notes were raised while tracing requirements to code, and are referenced from docs 01–04 and 07. Their status was re-verified against the codebase on 07 October 2026. **Severity:** H = compliance, security or money risk; M = functional gap or misleading behaviour; L = hygiene. **Status:** Resolved, Partly resolved or Open.

3.1 Summary

ID	Sev	Status	Finding (short)
E-01	H	Resolved	Voice campaign route bypassed the contact policy
E-02	M	Open (mitigated)	<code>voiceService.autoCall</code> does not itself refuse DNC customers
E-03	M	Partly resolved	Motorbike plates rejected; lead premium always uses the car tariff
E-04	M	Open	Sales channel <code>zalo</code> is never set; customer quotes hardcoded to <code>vetc_app</code>
E-05	M	Partly resolved	Business thresholds hardcoded in code
E-06	M	Resolved	MOTOR_PD inspection prerequisite not enforced
E-07	M	Resolved	Company-owned vehicles entered consumer journeys
E-08	L	Open	NBA <code>hot_call</code> ignores marketing consent
E-09	M	Open	Retention only partly executed (no anonymisation or archival)
E-10	M	Partly resolved	Benefits and wording promised features that do not exist
E-11	M	Partly resolved	Partner API: owner type, enumeration and data-poisoning risk
E-12	L	Resolved	Usable-data KPI used a DQ-score proxy
E-13	L	Open	<code>copy_guard.appliesTo</code> is ignored
E-14	L	Open	<code>wallet_topup</code> trigger uses a future-tense template for lapsed vehicles
E-15	M	Partly resolved	Token revocation and rate limiting held in memory per replica
E-16	L	Open	Insights do not scale to 6 M records
E-17	L	Open	Inconsistent phone masking in direct handoffs
E-18	M	Resolved	ABAC region checks missing on lineage, expiry correction and handoff detail
E-19	H	Resolved	DSAR erasure did not scrub handoffs, claims and voice data
E-20	M	Resolved	Quality gates (tests, lint, container, manifests) not yet runnable

ID	Sev	Status	Finding (short)
E-21	L	Open	Rule simulation limited to four rule kinds
E-22	H	Resolved	Staff-initiated purchase debited the customer's wallet

3.2 Resolved notes

- **E-01.** `POST /api/voice/campaign` now calls `voiceService.canCall`, which applies `canContact` (consent, DNC, 08:00–20:00 window, weekly call cap) per lead and reports skipped reasons.
- **E-06.** Purchase of `MOTOR_PD` is blocked until an inspection is recorded with `POST /api/quotes/:id/inspection` (`salesService.js`).
- **E-07.** Every consumer audience in `journeys.json` excludes `ownerType = company`.
- **E-12.** `profilesWithUsableData` now counts profiles without an open `reliable_expiry` issue, i.e. against `usableExpiryConfidence` (`insightsService.js#overview`).
- **E-18.** Lineage and expiry correction call the regional ABAC check, and `agent_own_handoffs` includes a region condition (`abac.json`).
- **E-19.** Erasure scrubs handoffs, claim text and voice signals, and invalidates the subject's sessions (`customerService.js#erase`).
- **E-20.** 238 automated tests, coverage gate, `eslint.config.js`, `Dockerfile`, Kubernetes manifests (`deploy/k8s`) and CI (`.github/workflows/ci.yml`) exist; referenced architecture documents and ADRs exist.
- **E-22.** `POST /api/orders` and permission `policy:issue` were removed. Staff send quotes with `POST /api/quotes/:id/send`; only the customer pays (`POST /api/customer/orders`).

3.3 Open and partly resolved notes

ID	Remaining issue	Evidence	Recommendation
E-02	<code>autoCall</code> relies on its callers; all current callers check <code>canCall</code> first	<code>voiceService.js</code>	Guard DNC and consent inside <code>autoCall</code>
E-03	Plate formats fixed; lead premium still uses <code>tariff.tnds_car</code> for every vehicle	<code>leadService.js#tndsPremium</code>	Choose the tariff by vehicle class
E-04	Zalo mini-app sales cannot be attributed or restricted	<code>routes.js</code> (<code>POST /api/customer/quotes</code>)	Derive the channel from the customer token
E-05	Remaining literals: <code>needsConfirmation</code> < 0.75, voice bot <code>expiryKnown</code> ≥ 0.5, bot +365 days, stuck-order 1 h, DQ completeness denominator	<code>customerService.js</code> , <code>voicebot.js</code> , <code>voiceService.js</code> , <code>opsService.js</code> , <code>enrichment.js</code>	Move into <code>service_levels</code> or <code>enrichment</code>
E-08	NBA can recommend a call that the contact policy then blocks	<code>nba.json</code> (<code>hot_call</code>)	Add <code>consent.marketing</code> to <code>hot_call</code>
E-09	Profile anonymisation and archival are reported as "executed by archival pipeline", which does not exist	<code>opsService.js#applyRetention</code>	Implement, or show as manual in the ops UI
E-10	<code>auto_renew</code> and <code>fleet_dashboard</code> are now staff-only (<code>available: false</code>); the summary of <code>POST /api/customer/expiry</code> still says "earns points"	<code>routes.js</code> , <code>docs/api/openapi.json</code>	Remove the points wording until loyalty is approved

ID	Remaining issue	Evidence	Recommendation
E-11	ownerType accepted and key scopes enforced; any active partner can still quote any plate and supply expiry evidence	routes.js (POST /api/partner/v1/quotes)	Rate-limit per key; flag conflicting partner evidence as DQ issues
E-13	Guard runs on all customer copy (conservative), but the configuration suggests otherwise	copy_guard.json, contactPolicy.js#checkCopy	Document "all copy" or implement scoping
E-14	Lapsed vehicles can receive "expires on {{expiry}}" with a past date	triggers.json	Use lapsed_notice when days < 0
E-15	Password and role changes revoke tokens on all replicas; logout revocation and rate limiting remain per replica	identityService.js, security.js	Shared store, or enforce at the gateway/WAF
E-16	adoption() reads the latest 5,000 audit entries; overview() pages through all orders	insightsService.js	SQL aggregates or materialised views
E-17	createDirectHandoff builds its own phone mask	voiceService.js	Use shared/util.js#maskPhone
E-21	No simulation for tariffs, contact policy or triggers	routes.js (simulate)	Add quote simulation and canContact dry-runs